Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified | DTE Code No: 3423 | Recognized under Section 2(f) of the UGC Act 1956 | Minority Status (Hindi Linguistic)

DEPARTMENT OF COMPUTER ENGINEERING

CML610 Cloud Computing Lab

Sixth Semester, 2025-2026 (Even Semester)

Name of Student : Ayan Parkar

Roll No. : 48

Division : A

Batch : A2

Day / Session :

Venue :

Experiment No. : 7

Title of Experiment : To study and Implement Database as a Service on SQL databases like AWS RDS

Date of Conduction :

Date of Submission :

Particulars	Max. Marks	Marks Obtained
Preparedness and Efforts(PE)	**3**	
Knowledge of tools(KT)	**3**	
Debugging and results(DR)	**3**	
Documentation(DN)	**3**	
Punctuality & Lab Ethics(PL)	**3**	
Total	**15**	

Grades – Meet Expectations (3 Marks), Moderate Expectations (2 Marks), Below Expectations (1 Mark)

Checked and Verified by

Name of Faculty : Pravin Jangid

Signature :

Date :

Experiment No 7

Aim: To study and Implement Database as a Service on SQL databases like AWS RDS

Theory:

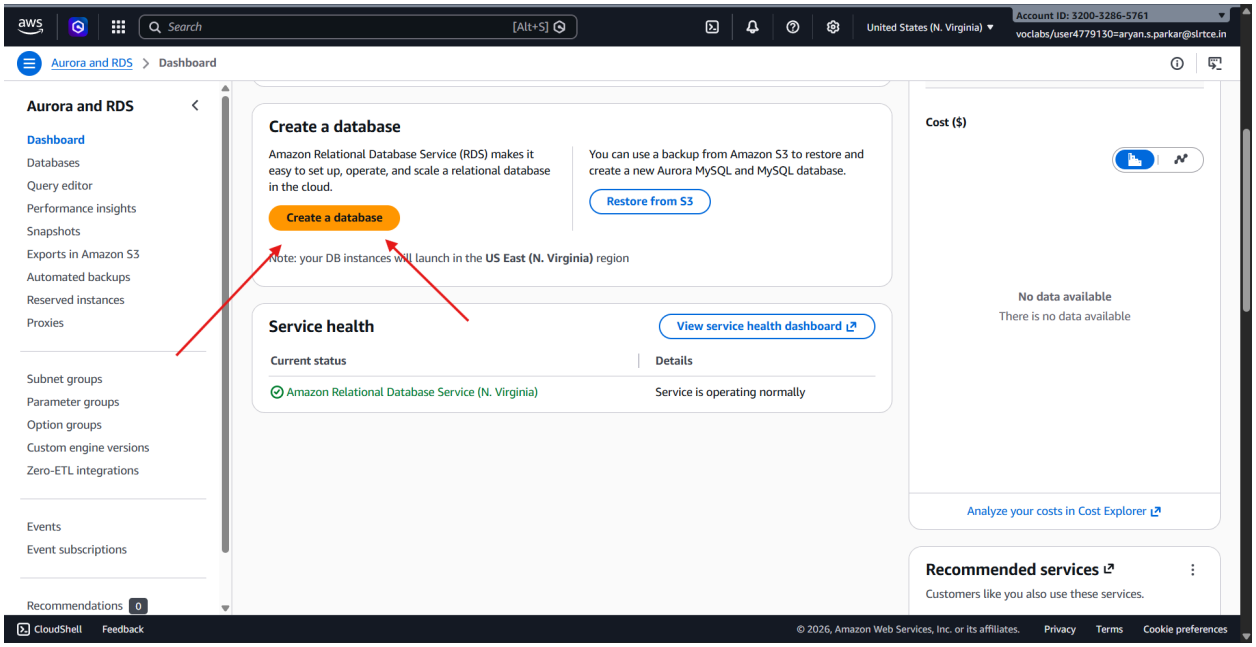
Aspect	Advantages	Limitations
Administration	Automates hardware provisioning, database setup, software patching, and system maintenance	Lacks underlying OS or root/SSH access, preventing the installation of third-party software or unsupported custom extensions
High Availability	Supports Multi-AZ deployments that provide synchronous data replication and automatic failover in case of hardware failure	Multi-AZ configurations double the underlying infrastructure costs, and the failover process can still cause brief latency spikes
Scalabiltiy	Allows easy horizontal scaling for read-heavy workloads using Read Replicas, and supports dynamic storage auto-scaling	Vertical scaling (changing instance types to increase compute or RAM) requires a brief period of downtime while the database reboots
Backup and Recovery	Features automated daily backups and snapshots, enabling Point-in-Time Recovery (PITR) with a retention period of up to 35 days	Heavy write workloads during snapshot creation can slightly degrade performance, and extended backup retention incurs additional storage costs
Security	Integrates natively with AWS Key Management Service (KMS) for at-rest encryption, VPC for network isolation, and IAM for secure access control	Customers must rely entirely on the AWS release schedule to roll out engine version updates and critical security patches
Cost Efficiency	Reduces operational overhead and the need for dedicated database administrators, lowering the overall Total Cost of Ownership (TCO)	Hourly compute and storage rates are notably higher than self-hosting the exact same database engine directly on Amazon EC2 instances

Activity:

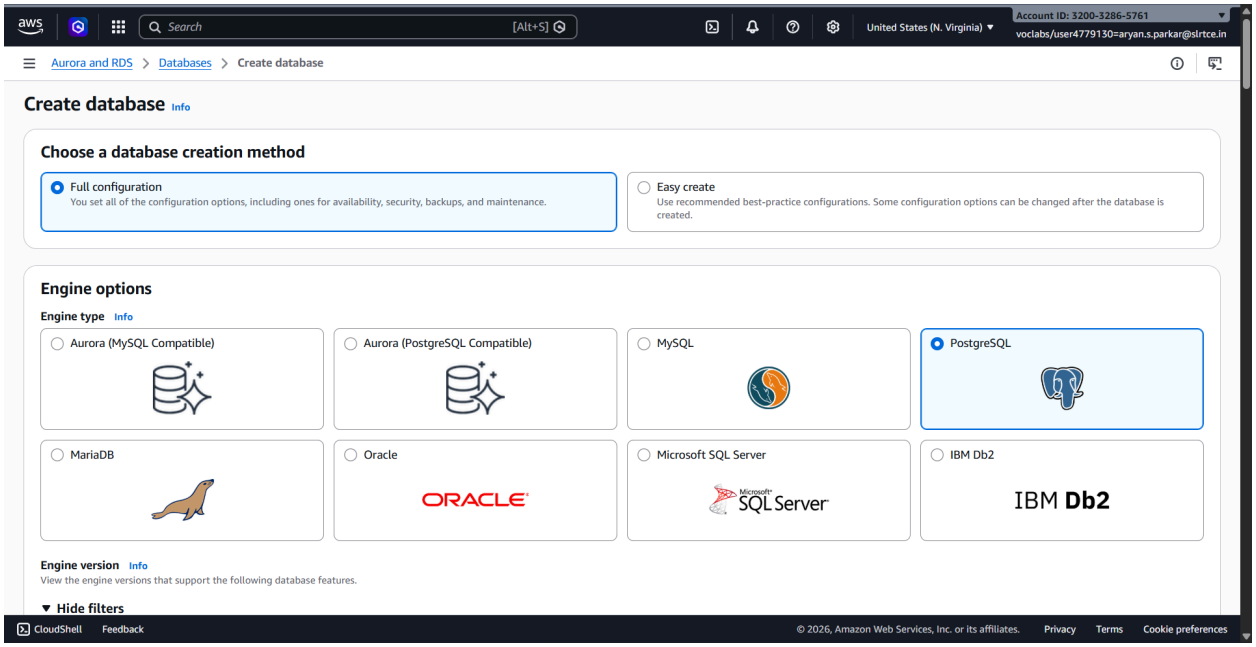
<u>Demonstrate the CRUD operations on different SQL and NoSQL databases running on cloud</u>

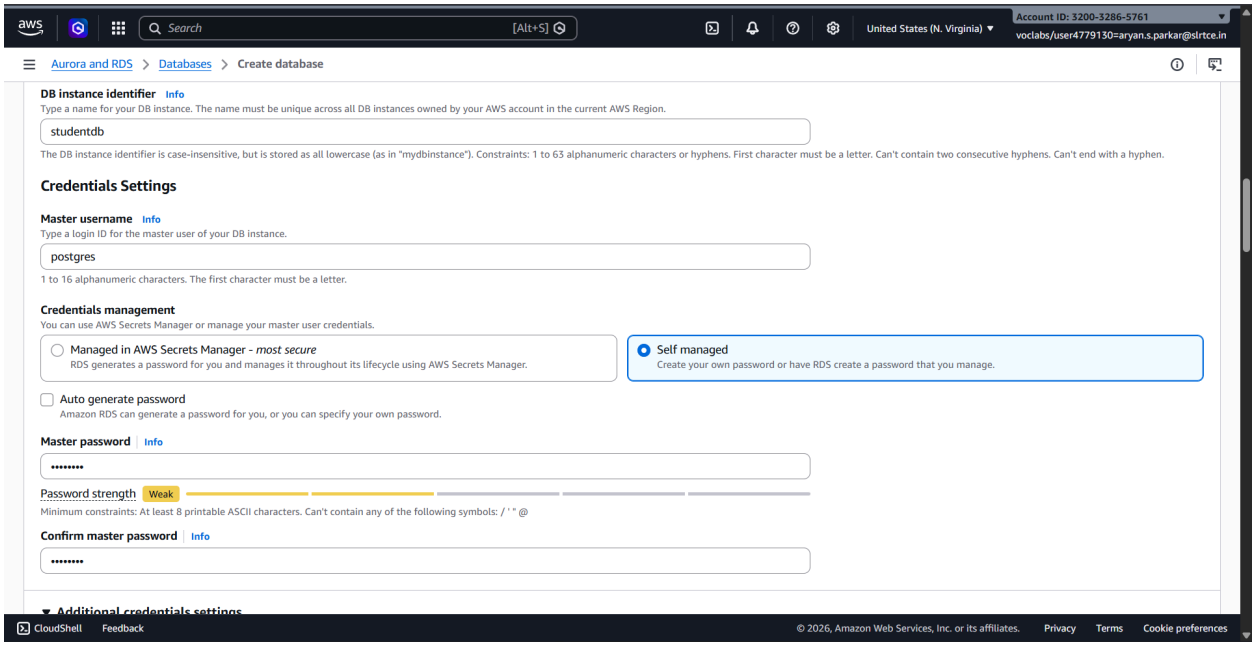
Step 1: Log in to AWS Console

Step 2: Search for RDS, click on Create Database



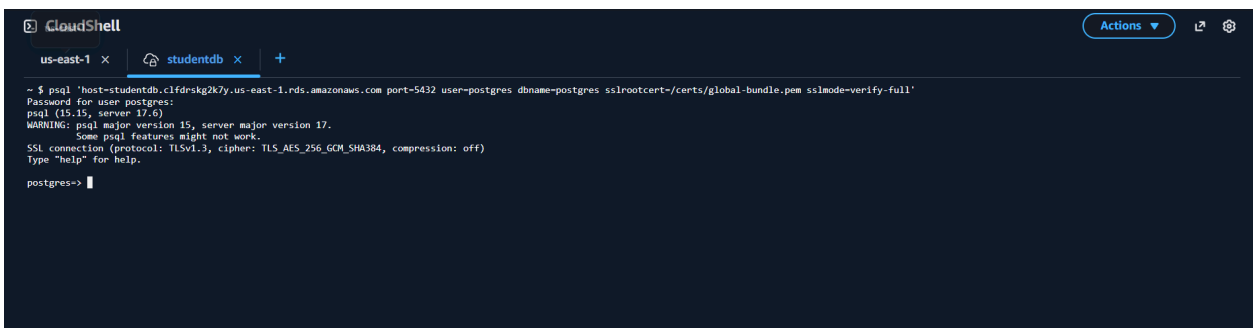
Step 3: Configure your database

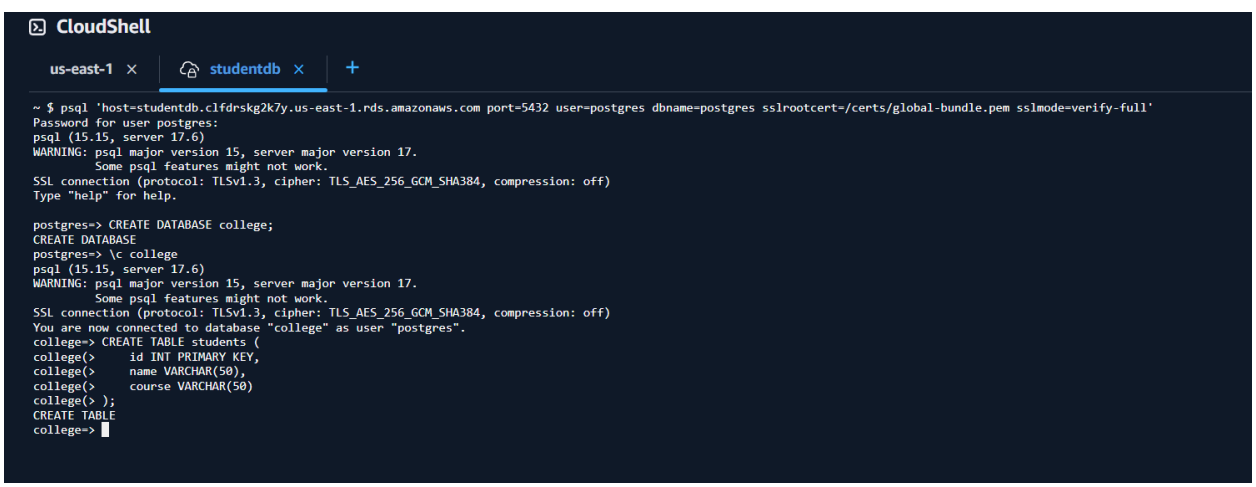




Step 4: Select on the All Access button and Create the Database

Step 5: Connect to the database using CloudShell





```
CREATE TABLE
college=> INSERT INTO students VALUES (1, 'Ayan', 'Computer Science');
INSERT 0 1
college=> INSERT INTO students VALUES (2, 'Rahul', 'Data Science');
INSERT 0 1
college=> INSERT INTO students VALUES (3, 'Neha', 'Cyber Security');
INSERT 0 1
college=> SELECT * FROM students;
 id | name  |      course
----+-------+-----------------
  1 | Ayan  | Computer Science
  2 | Rahul | Data Science
  3 | Neha  | Cyber Security
(3 rows)

college=>
```

```
INSERT 0 1
college=> SELECT * FROM students;
 id | name  |      course
----+-------+-----------------
  1 | Ayan  | Computer Science
  2 | Rahul | Data Science
  3 | Neha  | Cyber Security
(3 rows)

college=> UPDATE students
college-> SET course = 'Artificial Intelligence'
college-> WHERE id = 2;
UPDATE 1
college=> SELECT * FROM students;
 id | name  |         course
----+-------+------------------------
  1 | Ayan  | Computer Science
  3 | Neha  | Cyber Security
  2 | Rahul | Artificial Intelligence
(3 rows)

college=>
```

```
college-> SET course = 'Artificial Intelligence'
college-> WHERE id = 2;
UPDATE 1
college=> SELECT * FROM students;
 id | name  |         course
----+-------+------------------------
  1 | Ayan  | Computer Science
  3 | Neha  | Cyber Security
  2 | Rahul | Artificial Intelligence
(3 rows)

college=> DELETE FROM students
college-> WHERE id = 3;
DELETE 1
college=> SELECT * FROM students;
 id | name  |         course
----+-------+------------------------
  1 | Ayan  | Computer Science
  2 | Rahul | Artificial Intelligence
(2 rows)

college=>
```

Conclusion:

Hence, we successfully created a SQL database using Amazon RDS and performed CRUD (Create, Read, Update, Delete) operations on the students table.